

Parmeet Singh Granthi

granthi.parmeet@gmail.com | +91 7420004161 | parmeetsingh.com | [LinkedIn](#)

Professional Summary

Agentic AI & ML Engineer specializing in **multi-agent systems**, **RAG pipelines**, and **LLM/NLP applications**. Experienced in designing and deploying intelligent workflows using **LangChain**, **LangGraph**, **CrewAI**, and **LangSmith**, with strong expertise in **Python**, **PyTorch**, **Generative AI**, **Prompt Engineering**, and **fine-tuning LLMs**. Skilled in building scalable AI solutions with **vector databases (FAISS, Pinecone)** and deploying them through modern engineering stacks including **FastAPI**, **Docker**, **PostgreSQL**, **MLOps** and **cloud platforms (AWS/GCP)**. Passionate about translating AI research into **real-world applications**, with a focus on **safe, ethical, and well-guarded AI systems** through **evaluation, monitoring, and controlled deployment**.

Core Skills

- **Agentic AI & RAG:** RAG, LangChain, LangGraph, CrewAI, LangSmith (Evaluation & Monitoring)
- **Vector Databases & Retrieval:** FAISS, Pinecone, Weaviate, Document Loaders, Text Splitting, Retrievers
- **AI & ML:** Python, PyTorch, Generative AI, NLP, LLMs, Prompt Engineering, Fine-Tuning, Chatbots, MLOps
- **Build & Deploy:** FastAPI, Docker, Streamlit, CI/CD, AWS/GCP, PostgreSQL, Pydantic, Integrations, Security
- **Data, Tools & Methods:** NumPy, Pandas, Agile, SDLC, Git, Jira

Experience

Kiran Engineering Works – AI & Software Engineer

Sep 2017 – Present · Chhatrapati Sambhajinagar, India (formerly Aurangabad)

- Built **ML anomaly-detection dashboards** on **time-series sensor data** for real-time health monitoring.
- Implemented **RAG assistants (LangChain, FAISS, OpenAI)** for internal ops, reducing lookup time by **70%**.
- Built **multi-agent LLM workflows (LangSmith)** for compliance automation, saving **50+ hrs/month**.
- Automated finance and HR workflows using **OCR, embeddings, and controlled LLM generation**.
- Trained **predictive ML models** on sensor and operational data, reducing downtime by **30%**.
- Deployed **scheduled ML inference pipelines** with logging, improving system uptime by **35%**.
- Integrated **LLM-based retrieval and response generation** into CRM, improving response time by **60%**.
- Deployed **production AI systems** using **Python, FastAPI, Docker, CI/CD, MLOps on AWS/GCP**.

Independent Projects / Portfolio – Agentic AI & RAG Engineer

Live at parmeetsingh.com

- Featured Projects: [YouTube RAG](#) · [NetZero Advisor](#) · [Research Agent](#) · [Legal Document Analyzer](#)
- **YouTube RAG:** Turns YouTube videos into a **searchable knowledge** base with **transcript grounded Q and A**.
- **NetZero Advisor:** AI advisor for **carbon credits and sustainability insights** using retrieval and reasoning.
- **Research Agent:** Automates **multi source research** by collecting and summarizing relevant information.
- **Legal Document Analyzer:** Answers questions from **long legal documents** using **structured retrieval**.

LTI – Larsen & Toubro Infotech Ltd. – Software Engineer

Apr 2015 – Jul 2017 · Pune, India

- **Assisted** in development and maintenance of **enterprise applications**.
- Worked with **Python** for scripting tasks and building/supporting **REST APIs**.
- Contributed to **testing, debugging, and deployment** processes.
- Gained strong exposure to **Agile workflows** and **SDLC practices**.

Education

- **B.E. Computer Science Engineering** – PESCOE, Aurangabad (First Class with Distinction, 2010–2014)